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Present:

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Start of meeting

1 Progress and Pre-Meeting Notes

Progress was reviewed on the preprocessing pipeline and feature engineering steps established in the previous meeting.

2 Data Preprocessing and Windowing

5 2.1 Inter-Pass Boundary Correction

- **First Window Removal per Pass Type:** The first windowed sample of each new pass type segment must be discarded, as it inherently belongs to the previous pass regime. This transition window produces a hybrid feature vector that is not representative of either pass type and introduces noise into the training data.

10 3 Feature Engineering and Future Covariates

3.1 Shifting Columns into the Future

- **Future Covariate Design:** A key design decision was discussed regarding which columns should be shifted by one step into the future (i.e., t comes back: (3 / 3) present 1) to serve as known future covariates for the TFT model. The following candidates were identified:
 - 15 ▶ **Pass Type:** The pass type at step t comes back: (3 / 3) present 1 is deterministically known from the CNC program in advance and should be shifted accordingly.
 - ▶ **Spindle Speed and Feed Rate:** Similarly, operational parameters such as spindle speed and axis velocity are pre-programmed and fully known before execution, making them valid future covariates.
- **Justification:** Since the CNC machining program is planned ahead of time, virtually all operational context parameters at t comes back: (3 / 3) present 1 are known at decision time t , with the sole

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exception of the spindle torque (y), which is the prediction target. This property is a core architectural advantage of the TFT model over standard LSTM approaches.

Meeting Adjourned

25 *Notes reformulated with AI assistance*

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